

ESP32 Technical Reference

Week 7 Portfolio Reference

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PROJECT DOCUMENTATION & HARDWARE ARCHITECTURE

Target: ESP32-WROOM-32D (30-pin)

Project Documentation Notice: This document is created specifically for the Week 7 IoT, Connectivity & RTOS portfolio project. It details the exact hardware implementation, GPIO pin assignments, FreeRTOS dual-core multitasking model, and peripheral mapping utilized in Tasks 1 through 5.

1. System Architecture & Processor Specs

The core controller used throughout Week 7 is the ESP32 Dev Module, featuring an Espressif ESP-WROOM-32 module with dual Tensilica Xtensa 32-bit LX6 microprocessors running at 240 MHz. In Tasks 4 and 5, the dual-core capability is leveraged to split time-critical local automation (Core 0) and network/cloud synchronization (Core 1).

2. Hardware Pinout & Project GPIO Map

GPIO	Pin Name	Peripheral / Function	Connected Component in Project	Voltage
GPIO 2	D2	Internal LED / Output	System Diagnostic & Mode Status LED	3.3V Logic
GPIO 4	D4	ADC2_CH0 / Touch 0 / GPIO	DHT11 Digital One-Wire Data Bus	3.3V Logic (10k Pull-up)
GPIO 18	D18	VSPI CLK / GPIO Output	5V Relay Module Input (Optocoupled)	5V VCC / 3.3V IN
GPIO 34	D34	ADC1_CH6 (Input Only)	LDR Analog Voltage Divider Output	Analog 0–3.3V (12-bit ADC)
TX0 / RX0	GPIO 1 / 3	Hardware UART0	USB-Serial Flashing & Real-time Telemetry Monitor	115200 Baud

3. FreeRTOS Dual-Core Multitasking Model

In Task 5, standard single-thread loop execution is replaced with FreeRTOS preemptive task scheduling:

- Core 0 (Task Automation):** Assigned 4096 bytes stack depth. Executes with Priority 2. Reads DHT11 digital signals and LDR analog voltages. Performs local threshold evaluation ($LDR < 1200$) and controls the relay autonomously in sub-millisecond response times.
- Core 1 (Task Cloud Sync):** Assigned 8192 bytes stack depth. Executes with Priority 1. Manages Wi-Fi TCP/IP stack and non-blocking asynchronous Firebase Realtime Database updates.

4. Power & Electrical Considerations

The board operates from a 5V USB connection regulated by an on-board AMS1117 3.3V LDO. Transmit bursts during Wi-Fi 802.11 b/g/n operation draw up to 240 mA; an external 100uF decoupling capacitor across the 3.3V power rails ensures stable ADC readings without brownout resets.